

RallyClip: Segment-Level Evaluation Metrics and an End-to-End Loss Function

(spec only — not implemented yet)

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Motivation

The product objective is “good clips”: a predicted point should start a little before the actual point and end a little after it. We care about **absolute boundary distance in seconds**, not IoU’s relative scaling — a 0.5s late start is equally bad on a 3s point and a 30s point. IoU is used only for matching and as a coarse gate for the lower tiers. This complements (does not replace) the existing frame-level metrics (balanced accuracy, frame F1, AUROC).

1 Definitions

Ground-truth (GT) point $G = [g_s, g_e]$ and predicted segment $P = [p_s, p_e]$, in seconds. For a matched pair, define signed boundary errors

$$e_s = p_s - g_s, \quad e_e = p_e - g_e.$$

Negative e_s (early start) and positive e_e (late end) add *extra footage* around the point: the **outside** direction, mildly penalized. Positive e_s and negative e_e *cut into* the point: the **inside** direction, strictly penalized.

2 Evaluation metric: the 6-bin proportion

2.1 Flow

1. **IoU cross-matrix** between every prediction and every GT point.
2. **Falses.** A prediction with $\max_G \text{IoU}(P, G) < 0.1$ is a *false positive*; a GT point with $\max_P \text{IoU}(P, G) < 0.1$ is a *false negative*. Both are removed before matching.
3. **Resolve to a 1-to-1 mapping.** Remaining entities may form many-to-one mappings (two predictions on one GT = split; one prediction spanning two GTs = merge). When this happens the outcome is definitely not a correct classification, regardless of resolution. Heuristic:
 - (a) Match pairs with $\text{IoU} \geq 0.1$ **greedily by descending IoU**, each prediction and each GT used at most once.
 - (b) Leftover *predictions* (duplicates/fragments from a split) $\rightarrow$ counted as **false positives** (an extra clip the user must delete).
 - (c) Leftover *GT points* (losing side of a merge) $\rightarrow$ **false negatives** (that point gets no clip of its own).

- (d) The winning pair is tiered normally: its boundary errors are large by construction, so the tolerances naturally demote it out of Good/Decent.
4. **Tier the matched pairs** (each pair gets the best tier it satisfies).

2.2 Tiers

#	Tier	Condition
1	Good	$-0.5 \leq e_s \leq +0.2$ and $-0.2 \leq e_e \leq +0.5$
2	Decent	$-1.5 \leq e_s \leq +0.5$ and $-0.5 \leq e_e \leq +1.5$ (and not Good)
3	Recognized, bad segmentation	$\text{IoU} \geq 0.5$, outside Decent tolerances
4	Poor point recognition	$0.1 \leq \text{IoU} < 0.5$
5	False positive	prediction, $\max_G \text{IoU} < 0.1$ (or unmatched leftover)
6	False negative	GT point, $\max_P \text{IoU} < 0.1$ (or unmatched leftover)

Tolerance convention per boundary: *outside* / *inside*. Good = 0.5 s outside, 0.2 s inside; Decent = 1.5 s outside, 0.5 s inside.

2.3 Main description

Every prediction and every GT point lands in exactly one group occurrence (a matched pair counts once). With $n_{\text{events}} = n_{\text{matched}} + n_{\text{FP}} + n_{\text{FN}}$, report

$$\text{share}(g) = \frac{\text{count}(g)}{n_{\text{events}}}, \quad g \in \{1, \dots, 6\}.$$

Ideally everything is in Good, maybe Decent. **Optimization order** (worst first): (1) minimize falses (FP/FN), (2) poor point recognition, (3) bad segmentation, (4) move Decent $\rightarrow$ Good.

2.4 Secondary metrics

False positives per hour (FPs have no GT denominator); mean/median/p90 of signed e_s, e_e over matched pairs (systematic early/late bias informs postprocess tuning); per-video breakdown. If splits/merges are frequent, add diagnostic counters $n_{\text{split}}, n_{\text{merge}}$ without changing the taxonomy.

3 Loss function (end-to-end segment prediction)

For a future e2e model that predicts segments directly, replacing hysteresis filtering. Kept strictly separate from the metric: the *metric* is thresholded and interpretable; the *loss* is a continuous relaxation. It must **not** replicate the metric’s bin thresholds — hard thresholds create zero-gradient plateaus inside bins and cliffs at edges. The only intentional flat region is the Good zone itself. The 0.1 IoU cutoff exists *only in the metric*: in the loss, a 0.09-IoU near-miss naturally incurs less loss than a zero-overlap miss, with no special casing.

3.1 Prerequisite: segment-predicting head

The model must output tuples $(\hat{c}, \hat{p}_s, \hat{p}_e)$ — confidence, start, end. Two candidate designs (decision deferred): **(a)** DETR-style set prediction with K learnable queries, NMS-free; **(b)** anchor-free per-frame regression (ActionFormer-style): each frame predicts (pointness_{*t*}, d_t^{start} , d_t^{end}) on top of the

existing BiLSTM. The loss below is written in form (a); it translates to (b) with the boundary hinge applied to offset regressions of in-point frames and confidence terms becoming per-frame focal loss.

3.2 Matching (training time)

Hungarian matching between the K predictions and the n GT points with cost $-w_c\hat{c}$ +boundary cost+IoU cost. With $K \geq n_{\text{gt}}$, **every GT point is matched**. This is the mechanism that makes false negatives punishable: a missed point is not a separate loss term but a matched pair with large boundary/IoU loss plus a confidence term pulling that slot up — the gradient always has a prediction to flow into.

3.3 Component 1: boundary loss — asymmetric two-knee hinge

Zero loss inside the Good zone, slope λ_1 until the Decent edge, slope λ_2 beyond. For one boundary with Good zone $[-g_{\text{out}}, +g_{\text{in}}]$ and Decent zone $[-d_{\text{out}}, +d_{\text{in}}]$, define the one-sided violations

$$u = \max(-e - g_{\text{out}}, 0) \quad (\text{outside}), \quad v = \max(e - g_{\text{in}}, 0) \quad (\text{inside}),$$

and

$$L_b(e) = \underbrace{\lambda_1 \min(u, \Delta_{\text{out}}) + \lambda_2 \max(u - \Delta_{\text{out}}, 0)}_{\text{outside branch}} + \underbrace{\mu_1 \min(v, \Delta_{\text{in}}) + \mu_2 \max(v - \Delta_{\text{in}}, 0)}_{\text{inside branch}}$$

where $\Delta_{\text{out}} = d_{\text{out}} - g_{\text{out}}$ and $\Delta_{\text{in}} = d_{\text{in}} - g_{\text{in}}$. For the start boundary $g_{\text{out}}=0.5$, $g_{\text{in}}=0.2$, $d_{\text{out}}=1.5$, $d_{\text{in}}=0.5$; the end boundary is mirrored. Constraints: $\lambda_1 < \lambda_2$, $\mu_1 < \mu_2$, and $\mu \geq \lambda$ (cutting off part of the point is worse than extra footage). L_b is continuous, convex, piecewise-linear; kinks are ReLU-like and SGD-safe (optionally Huber-smoothed). Total per pair: $L_{\text{boundary}} = L_b(e_s) + L_b(e_e)$.

3.4 Component 2: IoU — 1D DIoU

$$L_{\text{IoU}} = 1 - \text{DIoU}(P, G), \quad \text{DIoU} = \text{IoU} - \left(\frac{|c_P - c_G|}{\ell_{\text{enclose}}} \right)^2,$$

with c_P, c_G the segment centers and ℓ_{enclose} the length of the smallest interval containing both. Plain $1 - \text{IoU}$ has **zero gradient at zero overlap** (a prediction 1 s away and 50 s away both score $\text{IoU} = 0$) — it stops pointing in the right direction exactly where it matters most. The center-distance term keeps a nonzero gradient toward the GT at any distance and yields the graded 0-vs-0.09 behavior automatically. Weight β small: the hinges already carry the absolute-time objective; DIoU covers the far-miss regime.

3.5 Component 3: confidence

Focal/BCE on each slot's confidence $\hat{c}$: matched slots target 1 with weight w_{pos} (set high — an unconfident matched slot on a real point is an incipient **false negative**, the biggest sin); unmatched slots target 0 with weight w_{neg} — this is the **false-positive** penalty. Unmatched predictions get no boundary gradient (nothing legitimate to regress toward); confidence suppression is the correct and only gradient.

3.6 Total

$$L = \sum_{\text{matched}} \left[\alpha L_{\text{boundary}} + \beta L_{\text{IoU}} + w_{\text{pos}} \text{FL}(\hat{c}, 1) \right] + \sum_{\text{unmatched}} w_{\text{neg}} \text{FL}(\hat{c}, 0).$$

Weight ordering mirrors the metric’s optimization order: $w_{\text{pos}} (\text{FN}) \geq w_{\text{neg}} (\text{FP}) > \beta$, with the hinge slopes (λ, μ) shaping the segmentation-quality tail.

3.7 Gradient sanity checklist

- No zero-overlap dead zone: DIOU distance term and boundary hinges give gradients at arbitrary distance.
- No bin cliffs: every threshold in the loss is a hinge knee (continuous), never a step function.
- Intentional flat region: exactly the Good zone, nowhere else (no pressure toward the GT center once inside — any position in the zone is equally acceptable).
- FN gradient path: forced matching $\Rightarrow$ every GT injects gradient into some slot.
- FP gradient path: confidence suppression on unmatched slots.

3.8 Open implementation questions

Head choice (set prediction vs. per-frame regression) and K ; window truncation (20 s training sequences cut points at window edges — train on full videos via the BiLSTM, or mask/clip boundary targets at edges); exact coefficients $(\lambda_1, \lambda_2, \mu_1, \mu_2, \alpha, \beta, w_{\text{pos}}, w_{\text{neg}})$, tuned against the 6-bin metric; Huber smoothing of hinge knees and focal-loss γ .